

LLM-Driven Language Shift in Naturalistic Communication: Markers, Diffusion Pathways, and Social Consequences

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Abstract

The rapid diffusion of large language models (LLMs) has introduced new forms of human-AI interaction that are increasingly embedded in everyday communication practices. LLMs are now used not only to generate complete texts, but also to suggest alternative phrasings, provide outlines, revise tone, and support stylistic refinement (Bommasani et al., 2021; Ouyang et al., 2022). These uses create subtle yet pervasive opportunities for model-preferred lexical, stylistic, and discourse patterns to influence human language production.

While LLMs are commonly framed as tools for generating or assisting with text, emerging evidence and recent preprint findings suggest that exposure to LLM-generated language may also influence how humans write, speak, and evaluate communication (Anderson et al., 2025; Galpin et al., 2025). This project conducts a systematic review of empirical research examining how LLM exposure and use are associated with changes in human language and communication in naturalistic settings. Using a systematic review methodology aligned with PRISMA guidance (Page et al., 2021), the study synthesizes empirical research across linguistics, communication, human-computer interaction, and education to examine (a) which linguistic features show evidence of LLM-related change, (b) through which pathways LLM-influenced language may diffuse, and (c) what implications these shifts have for communication norms such as authenticity, credibility, and trust. At the time of submission, the review protocol, search strategy, screening procedures, and data extraction framework have been fully developed and implemented, with the core empirical corpus identified and synthesis underway. The report documents the conceptual framing, methods, and preliminary evidence mapping that position the project for completion of the full systematic synthesis. Unlike earlier writing technologies, LLMs produce fluent, contextually adaptive language at scale, which may prime users toward particular linguistic forms through repeated exposure.

Introduction

Large language models (LLMs) have transitioned rapidly from experimental systems to widely used tools for writing, revision, and interaction across digital platforms. Beyond generating

standalone content, contemporary LLMs are frequently used for autocomplete, paraphrasing, tone adjustment, and stylistic refinement, embedding them into everyday communicative workflows (Ouyang et al., 2022; Weidinger et al., 2021). As a result, people increasingly encounter LLM-generated or LLM-influenced language even in contexts where they are not directly interacting with these systems.

This widespread exposure raises important questions about whether, and how, LLM exposure influences how people write, speak, and evaluate communication. In psycholinguistics, decades of research demonstrate that repeated exposure to linguistic structures can shape subsequent production through mechanisms such as structural priming and implicit learning (Bock & Griffin, 2000; Chang et al., 2006; Pickering & Garrod, 2004). Separately, sociolinguistic research shows that linguistic innovations can diffuse through social networks and communities, often extending beyond direct adopters of a new form or technology. Studies of lexical innovation in social media and online discourse demonstrate that novel forms spread in ways consistent with diffusion and social coordination processes in language communities (Eisenstein, 2014; Paradowski, 2012).

Emerging empirical work suggests that characteristic LLM-associated patterns--such as shifts in lexical choice, increased formality or hedging, and recurrent discourse transitions--may be appearing in human-authored texts, and in some cases speech, following the public release of chat-based LLMs. For example, recent corpus-based studies report increased frequencies of lexical items that are overrepresented in LLM-generated text in online discussions after the release of systems such as ChatGPT, including among users who do not directly report using these tools themselves (Anderson et al., 2025; Galpin et al., 2025). However, this evidence is fragmented across domains and methods, with substantial variation in how “LLM-like” language is defined and measured, and in which outcomes are emphasized, from corpus-level frequency changes to perceptions of authenticity or credibility.

To date, there is no integrated synthesis focused on the relationship between LLM exposure and measurable changes in human language and communication across naturalistic settings. Existing reviews tend to focus on educational integrity, writing support tools, or conceptual and ethical considerations, rather than on empirical evidence of language change (e.g., Kasneci et al., 2023). This review addresses that gap through a systematic synthesis of empirical studies examining LLM-related language change. The work is guided by conceptual lenses from psycholinguistics (priming and implicit learning), sociolinguistics (diffusion and contagion), and communication research (norms, authenticity, and trust). The aim is to move beyond anecdotal claims about “people writing like LLMs” toward a structured understanding of what kinds of linguistic shifts have been observed, how they spread, and what social and normative implications they carry.

Conceptual Framework

The review is guided by five complementary conceptual lenses:

1. **Priming and implicit learning.** Repeated exposure to characteristic LLM output is treated as a potential source of priming and implicit learning, whereby lexical bundles,

hedging patterns, or discourse structures influence subsequent human language production without explicit intention. Psycholinguistic research has consistently shown that exposure to linguistic structures can produce persistent adaptation effects, even in the absence of conscious awareness (Chang et al., 2006).

2. **Diffusion and social contagion.** LLM-influenced language is conceptualized as potentially spreading through communities and platforms via social contagion, such that individuals who do not directly use LLMs may nonetheless adopt features that become prevalent in their communicative environments. Sociolinguistic research on language change demonstrates that innovations often diffuse unevenly across social networks, shaped by interaction patterns and identity signaling (Labov, 2007; Eisenstein, 2014).
3. **Norms, authenticity, and trust.** Changes in language use are examined in relation to emerging norms around authorship, authenticity, and credibility. As AI-assisted writing becomes more common, communities may renegotiate expectations about what constitutes “natural” or acceptable language, with potential consequences for perceived trustworthiness and expertise (Glikson & Woolley, 2020).
4. **Technological affordances and incentives.** LLM interfaces, platform reward structures, and institutional policies are treated as shaping adoption by enabling particular communicative behaviors and lowering the cost of producing certain linguistic styles. Building on affordance-based approaches to mediated interaction, this review treats technologies as offering structured possibilities and constraints that shape how people communicate in practice (Hutchby, 2001).
5. **Operationalization of “LLM-like” language.** Given the diversity of methods, the ways studies define and measure “LLM-like” language are treated as part of the object of analysis, informing interpretation and comparability.

These lenses structure the review’s research questions, data extraction, and synthesis, enabling integration of micro-level linguistic changes with broader social and normative implications.

Within this framework, the review attempts to address the following research questions:

1. How does exposure to and use of large language models (LLMs) influence human language and communication in naturalistic settings?
2. What measurable linguistic features (like vocabulary, tone, formality, hedging, discourse markers) show evidence of change associated with LLM adoption or exposure?
3. Through what pathways does LLM-influenced language spread (direct use vs. “secondhand” diffusion through communities and platforms)?
4. What downstream implications do these shifts have for communication norms such as authenticity, credibility, persuasion, and trust?

Methods

Review Design

This study follows a systematic review design aimed at cross-domain synthesis of empirical evidence on how LLM deployment is shaping human language and communication in naturalistic

contexts. The protocol specifies predefined research questions, inclusion and exclusion criteria, information sources, search strategies, screening procedures, data extraction categories, and a quality appraisal approach. The design accommodates both text-based and spoken modalities, public and institutional settings, and distinguishes between direct LLM use and secondhand diffusion through communities or platforms. The protocol was developed prior to full screening and has guided all subsequent stages of the review, following best practices for systematic review methodology to ensure transparency and reproducibility.

Information Sources and Search

The literature search was conducted across major bibliographic databases and preprint servers to capture relevant empirical work from multiple disciplines. Core databases include Scopus, Web of Science, PsycINFO, ERIC, ACM Digital Library, and IEEE Xplore. These sources were selected to ensure coverage of linguistics, psychology, education, communication, human–computer interaction, and computer science research. To complement peer-reviewed venues and capture rapidly emerging work, preprint servers such as arXiv, PsyArXiv, SocArXiv, and OSF Preprints were also searched. The primary search window covers publications from January 2022 onward, aligning with the period of widespread diffusion of chat-based LLMs, with a targeted supplemental search of earlier work on neural text generation for contextual background. Search strings combined terms related to LLMs, language and communication features, and naturalistic communication contexts. Reference lists of included studies were examined with backward and forward citation chasing for additional eligible works.

Eligibility Criteria

Studies were included if they met the following criteria: (a) they examine human language or communication outcomes in relation to exposure to or use of LLMs or closely related generative text models; (b) they employ empirical methods (quantitative, qualitative, or mixed); (c) they analyze naturalistic or ecologically valid settings (such as social media platforms, student or professional writing, public communication, or recorded speech) or realistic experimental tasks; and (d) they are reported in English. Outcomes may include lexical, stylistic, discourse, or pragmatic features, as well as perceptions or norms explicitly linked to LLM-influenced language.

Studies were excluded if they focus solely on model internals or technical performance without human language outcomes; address non-LLM chatbots without a clear link to LLM-related language change; are purely conceptual or opinion pieces without empirical data; or examine attitudes toward AI more generally without analyzing language use. Media articles, blog posts, and other non-scholarly sources were retained only as contextual material for the introduction and discussion. Theses and dissertations were noted but not systematically included, in order to maintain consistent standards of peer review and reporting.

Search Strategy

Search strings were developed to capture the intersection of LLM technologies, language or communication features, and naturalistic contexts. Terms referencing LLMs (e.g., "large language model*", "ChatGPT", "generative AI", "AI chatbot*") were combined with terms for language and

communication (e.g., "language use", "language change", "lexical choice", "writing style", "discourse marker*", "tone", "formality", "politeness", "hedging", "online communication") and with context terms reflecting settings such as "social media", "Reddit", "forum*", "student writing", "academic writing", "professional writing", or "podcast*". Search syntax was adapted to each database, and initial pilot searches were used to refine terms for adequate coverage and specificity. All searches were documented with the database, date, and full query strings. Backward and forward citation chasing were then applied to the pool of included studies to identify additional relevant work not captured by the initial database queries

Screening Procedures

Screening proceeded in two stages using the predefined eligibility criteria, following PRISMA-style guidance. In the first stage, titles and abstracts were screened to exclude clearly irrelevant records, such as purely technical LLM papers without human language outcomes or general AI attitude surveys. Each record was coded as "include", "exclude", or "uncertain" based on the abstract and, where necessary, indexing information. In the second stage, full texts were obtained for all "include" and "uncertain" records. These full texts were assessed in detail against the inclusion and exclusion criteria, with a specific reason logged for each exclusion (no empirical data, no LLM focus, no language outcome, non-English, etc.). Throughout both stages, decisions were guided by the protocol; borderline cases were evaluated and resolved based on how directly they addressed the research questions. The flow of records through identification, screening, eligibility assessment, and inclusion is being documented in a PRISMA-style diagram.

Data Extraction

A standardized extraction framework captures study design, communication context, exposure operationalization, linguistic features measured, diffusion pathways, and normative outcomes. Study quality is assessed using a light, design-sensitive rubric focusing on construct validity, analytical rigor, and transparency.

For each included study, the following information is recorded:

- Bibliographic details (authors, year, venue, discipline).
- Study design and methods (e.g., corpus analysis, pre-post design, survey, experiment, qualitative or mixed methods).
- Context and population (e.g., social media platform, educational course, workplace, public communication; students, general users, professionals).
- Language(s) analyzed and geographic or cultural setting, where applicable.
- Operationalization of LLM exposure or use (e.g., pre- vs post-LLM release time series, self-reported use in writing tasks, experimental assignment to use an LLM, inferred community-level exposure).
- Distinction between direct use of LLMs (e.g., copy-paste, assisted drafting or editing) and secondhand exposure via interaction with LLM-influenced content.
- Linguistic and communication outcomes, including levels of analysis (lexical, stylistic, discourse, pragmatic) and specific features measured (e.g., frequency of certain lexical items, measures of formality, prevalence of hedging or discourse markers, similarity to model output).

- Normative or evaluative outcomes, such as perceptions of authenticity, credibility, trustworthiness, or appropriateness of language suspected or known to be AI-influenced.
- Main findings relevant to each research question, noting direction and, where reported, magnitude of effects.
- Authors' reported limitations and any additional concerns noted during extraction.

Data extraction is being carried out using a standardized template, with initial entries cross-checked to ensure consistency. The coding scheme reflects the conceptual framework and is designed to support both descriptive mapping and thematic synthesis. At the time of submission, the core empirical corpus has been identified and fully screened, and structured data extraction is underway.

Quality appraisal

Given the methodological diversity of the included studies, a light, design-sensitive appraisal rubric is used rather than a single domain-specific tool. Each study is assessed along several dimensions: clarity and appropriateness of the research design; transparency and justification of sampling strategies and data sources; validity and reliability of measures for linguistic and normative outcomes; clarity of LLM exposure operationalization and its temporal or causal plausibility; and adequacy of analytical methods, including reporting of assumptions, robustness checks, or inter-coder reliability where relevant. Potential sources of bias and confounding (such as concurrent platform changes or broader cultural trends) are noted. Quality ratings are used to inform the interpretation of evidence in the synthesis--highlighting where conclusions rest on stronger versus weaker designs--but are not used to exclude studies unless they fail to provide interpretable data for any research question.

Use of LLMs within the review

LLMs are used cautiously and transparently within the review process. Assistance is limited to stylistic and editorial support, for example, suggesting revisions for clarity, adjusting tone, or restructuring sections of the narrative text. All protocol decisions, screening judgments, data extraction, coding, quality appraisal, and substantive interpretations are carried out by the researcher(s) and are not delegated to LLMs. LLMs are not used to make inclusion or exclusion decisions, to automatically extract data from articles without human verification, or to generate conclusions about the evidence base. This constraint is intended to preserve methodological rigor and accountability while still allowing limited use of LLM tools for improving presentation.

Preliminary Results

At the time of reporting, the multi-database and preprint server search has been completed, and the set of records has undergone full title/abstract and full-text screening. A core empirical corpus of studies meeting the inclusion criteria has been identified. Structured data extraction is in progress across this corpus, and an initial evidence map has been developed.

Initial evidence mapping indicates that the existing literature is:

- Concentrated in text-based communication, particularly social media, student writing, and scientific or professional prose.
- Less developed for spoken language, with only a small number of studies examining unscripted speech or conversational contexts.
- Methodologically diverse, including corpus analyses, pre-post designs, experiments, and surveys.
- Skewed toward direct or assisted LLM use, with comparatively fewer studies explicitly addressing secondhand diffusion.

Across studies, commonly examined linguistic features include lexical choice, stylistic formality, hedging, and discourse markers. A smaller subset of studies directly measures perceptions of authenticity, credibility, or trust, highlighting an important gap between linguistic change and social evaluation.

These patterns suggest emerging convergence on certain linguistic effects alongside substantial gaps in research documenting linguistic change and research directly linking those changes to shifts in communication norms and social evaluations.

Preliminary Discussion

Taken together, the reviewed studies identified so far suggest that LLM deployment is associated with detectable shifts in human language use, particularly in written communication where tools for drafting and polishing are most widespread. Evidence for lexical and stylistic changes appears in multiple settings, though the strength, direction, and generality of these effects vary by context, population, and methodological approach. Evidence for impacts on spoken language and for secondhand diffusion, where LLM-influenced language spreads to non-users via community norms, is more limited but points to important lines of future inquiry.

The diversity of operationalizations of "LLM-like" language complicates direct comparison across studies, underscoring the value of treating these operationalizations as part of the object of study. Normative implications, including concerns about authenticity, changing expectations of "natural" language, and the potential standardization or "flattening" of linguistic expression, are widely discussed but less often supported by systematic empirical measures. The ongoing synthesis will further differentiate robust patterns from tentative or context-specific findings and will connect micro-level linguistic shifts to broader questions about how AI integration is reshaping communication norms.

At the time of submission, this project has completed:

- development of a comprehensive conceptual framework,
- construction and execution of a multi-database search strategy,
- full screening and identification of the core empirical corpus,
- implementation of a structured data extraction and quality appraisal framework,
- preliminary evidence mapping across contexts, modalities, and methods.

Remaining work consists primarily of:

- completing full data extraction across all included studies,
- finalizing thematic synthesis aligned with the research questions,
- refining the discussion of implications, gaps, and future research directions.

The final synthesis will integrate these findings more fully, distinguishing robust patterns from tentative observations and situating linguistic change within broader sociotechnical dynamics. The completed review aims to provide a rigorous, cross-disciplinary synthesis of how LLMs are influencing human language and communication and to clarify where further empirical work is most urgently needed.

Conclusion

This review documents substantial progress toward a systematic synthesis of how large language models are shaping human language and communication. By integrating evidence across disciplines and settings, the project advances understanding of LLM-related language change while clearly identifying limitations in the current literature. The work is positioned to make a timely and rigorous contribution to ongoing debates about AI, language, and social communication norms.

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